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Savitzky-Golay split command-lag search

rotor median publish period [s]: 0.00499773025513
gimbal median publish period [s]: 0.00499868392944
initial lags [s]: {'rotor': 0.004997730255126953, 'gimbal': 0.004998683929443359}
smooth half-width multipliers: [4.0, 2.0, 1.0, 0.5]

smooth width=4: rotor=0.160213301s gimbal=0.0436244072s cost=835.912412975
    lag_gradient={'rotor': -0.37125098924535355, 'gimbal': -0.2741801445205372}
    FD=[{'direction': 'negative_gradient', 'scheme': 'central', 'step': 6.239460092292944e-05,
    'analytic_l2_norm': 367.09019295462537, 'finite_difference_l2_norm': 367.09019292787326,
    'difference_l2_norm': 7.511184941698178e-07, 'relative_error': 2.0461415439193186e-09},
    {'direction': 'rotor_delay_seconds', 'scheme': 'central', 'step': 6.239460092292944e-05,
    'analytic_l2_norm': 15.631528473575933, 'finite_difference_l2_norm': 15.631515660267201,
    'difference_l2_norm': 6.165725812492682e-05, 'relative_error': 3.9444164548050655e-06},
    {'direction': 'gimbal_delay_seconds', 'scheme': 'central', 'step': 6.239460092292944e-05,
    'analytic_l2_norm': 22.544146168964335, 'finite_difference_l2_norm': 22.54411602765875,
    'difference_l2_norm': 0.00012249386241070973, 'relative_error': 5.433510832153065e-06}]
smooth width=2: rotor=0.151430678s gimbal=0.0420252559s cost=835.791079508
    lag_gradient={'rotor': 0.18753882386417636, 'gimbal': 5.2034017550034095} FD=[{'direction':
    'negative_gradient', 'scheme': 'central', 'step': 6.229262571201071e-05, 'analytic_l2_norm':
     178.1032408639327, 'finite_difference_l2_norm': 178.13232290219426, 'difference_l2_norm':
    1.7532780820522618, 'relative_error': 0.009842560033391136}, {'direction':
    'rotor_delay_seconds', 'scheme': 'central', 'step': 6.229262571201071e-05,
    'analytic_l2_norm': 17.811204233012813, 'finite_difference_l2_norm': 17.811010258588563,
    'difference_l2_norm': 0.0004576803180179535, 'relative_error': 2.569620290859669e-05},
    {'direction': 'gimbal_delay_seconds', 'scheme': 'central', 'step': 6.229262571201071e-05,
    'analytic_l2_norm': 32.52236308172657, 'finite_difference_l2_norm': 32.575690584285745,
    'difference_l2_norm': 1.9796819257271798, 'relative_error': 0.06077175618441571}]
smooth width=1: rotor=0.152573009s gimbal=0.0464132479s cost=835.633940576
    lag_gradient={'rotor': -2.068139862316595, 'gimbal': -45.721760868763816} FD=[{'direction':
    'negative_gradient', 'scheme': 'central', 'step': 6.239291909571465e-05, 'analytic_l2_norm':
     74.98336637307587, 'finite_difference_l2_norm': 70.9162692634509, 'difference_l2_norm':
    19.359639229859003, 'relative_error': 0.2581857839448834}, {'direction':
    'rotor_delay_seconds', 'scheme': 'central', 'step': 6.239291909571465e-05,
    'analytic_l2_norm': 21.649400262935718, 'finite_difference_l2_norm': 21.648173683203368,
    'difference_l2_norm': 0.0031095398393705316, 'relative_error': 0.00014363168501688875},
    {'direction': 'gimbal_delay_seconds', 'scheme': 'central', 'step': 6.239291909571465e-05,
    'analytic_l2_norm': 67.39153360229147, 'finite_difference_l2_norm': 62.85812855189083,
    'difference_l2_norm': 19.460678259872417, 'relative_error': 0.2887703724731782}]
smooth width=0.5: rotor=0.152624524s gimbal=0.046265164s cost=835.597993708
    lag_gradient={'rotor': -21.859905506951055, 'gimbal': 184.1272456108536} FD=[{'direction':
    'negative_gradient', 'scheme': 'central', 'step': 6.239714190351619e-05, 'analytic_l2_norm':
     109.32026465246695, 'finite_difference_l2_norm': 108.5874735096851, 'difference_l2_norm':
    36.58884713391425, 'relative_error': 0.3346940958314683}, {'direction':
    'rotor_delay_seconds', 'scheme': 'central', 'step': 6.239714190351619e-05,
    'analytic_l2_norm': 28.50034701025338, 'finite_difference_l2_norm': 28.491854931492526,
    'difference_l2_norm': 0.016599980269232043, 'relative_error': 0.0005824483562694861},
    {'direction': 'gimbal_delay_seconds', 'scheme': 'central', 'step': 6.239714190351619e-05,
    'analytic_l2_norm': 109.5422500617485, 'finite_difference_l2_norm': 108.90520941012231,
    'difference_l2_norm': 36.88839907238366, 'relative_error': 0.3367504232530355}]

strict profile iterations: 1
iteration 1: best=(0.152624524376, 0.046265163997) cost=835.759211326
    rotor_range=(0.147626794121, 0.157622254631) gimbal_range=(0.041266480068, 0.051263847926)
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    expansions=[]

selected rotor lag 0.152624524376 s (30.5388 publish periods)
selected gimbal lag 0.046265163997 s (9.25547 publish periods)
selected objective cost 835.733074432
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